Decision Support Systems 180 (2024) 114190

Contents lists available at ScienceDirect

Decision Support Systems

journal homepage: www.elsevier.com/locate/dss

Distilling wisdom of crowds in online communities: A novel prediction market constructed with comment posters

Li Dong [a], Haichao Zheng [b,*], Liting Li [c], Chunyu Zhou [d]

[a] School of Management Science and Engineering, Southwestern University of Finance and Economics, Chengdu, China
[b] Financial Intelligence and Financial Engineering Key Laboratory of Sichuan Province, Southwestern University of Finance and Economics, Chengdu, China
[c] College of Economics and Management, Taiyuan University of Technology, Taiyuan, China
[d] Qube Research and Technologies, Hong Kong, China

ARTICLE INFO

Keywords:
Prediction market
Wisdom of crowds
Online community
Stock price movement prediction

ABSTRACT

Aggregating the wisdom of crowds from user-generated content in the online community can be valuable for decision-making. However, low-quality comments pose significant challenges for traditional wisdom extraction algorithms, such as prediction polls. Therefore, to extract the wisdom of online crowds effectively, we propose a novel artificial prediction market that can dynamically filter out low-quality comments using the market mechanism. Unlike traditional prediction markets where real human traders participate in contract trading, traders in the proposed market, referred to as artificial human traders, are constructed with posters' online comments. We constructed artificial human traders using the comment data from the Eastmoney stock forum. Then, we validated the effectiveness of the proposed market in predicting the price movements of the constituent stocks in the Shanghai Stock Exchange 50 Index. Experimental results demonstrate that the proposed market excels in extracting the wisdom of crowds from online communities for stock price movement predictions and conducting investment portfolios based on these prediction results. Additionally, to investigate the factors contributing to the superior performance of the proposed artificial prediction market, we compare it with the prediction poll, a well-known opinion aggregation algorithm. The results indicate that the artificial prediction market performs better in assessing the quality of comments compared with the prediction poll.

1. Introduction

User-generated content in online communities has served as a source of valuable insights. The value of this content is particularly evident in areas such as stock investments [1,2]. For instance, EastMoney, a prominent stock forum in China [3], facilitates investors exchanging investment experiences, connecting with other stock enthusiasts, and expressing their investment opinions. The abundant comments on these platforms offer substantial potential to harness crowd wisdom within online communities to formulate profitable stock investment strategies [2].

Effectively leveraging the collective wisdom embedded in user comments is crucial for investment decisions, yet it remains a formidable challenge [1]. This challenge primarily arises due to the prevalence of low-quality comments [4,5], which may jeopardize the effectiveness of opinion-mining methods [6]. Traditional methods usually use sentiment analysis, especially machine learning methods, to mine user opinions. However, machine learning models have shortcomings, such as inherent sensitivity to noisy data and outliers, computational intensity, and a complex "black box" nature [7]. Moreover, these methods generally are unable to effectively discern between high- and low-quality comments. While some voting methods (e.g., Soft Voting) can aggregate user opinions by assessing the dynamic qualities of posters' comments [6], assessing the quality of these comments is a significant challenge [8].

Prediction markets, an effective method of harnessing crowd wisdom to enhance decision-making [1], emerge as a promising solution in this context, enabling the aggregation of diverse opinions and predictions

* Corresponding author.
E-mail address: haichao@swufe.edu.cn (H. Zheng).

https://doi.org/10.1016/j.dss.2024.114190
Received 28 May 2023; Received in revised form 24 January 2024; Accepted 5 February 2024
Available online 9 February 2024

about future events through speculative trading of contracts. In prediction markets, participants (or traders) can buy or sell contracts with payoffs tied to specific future outcomes, such as "Candidate A will win the election" [1], and market prices fluctuate to reflect the aggregated beliefs of traders about the likelihood of event outcomes. An important characteristic of prediction markets is that traders who perform well thrive as the market progresses, while those who do not may face bankruptcy and eventually exit the market [9]. Prediction markets have showcased versatility across various domains, such as politics, finance, and sports [10]. For instance, in the Iowa Electronic Markets (https://iemweb.biz.uiowa.edu), participants can trade "win" contracts related to electoral outcomes [11].

This study, recognizing the value of prediction markets, proposed a novel artificial prediction market to integrate the wisdom of crowds within online communities. In our market, traders are neither individuals from the real world nor trained machine traders. Instead, they are traders constructed based on online comments from posters, referred to as artificial human traders. These artificial human traders, or traders, are armed with trading beliefs extracted from online comments and trading strategies developed through economic modeling methods. They speculate on the outcomes of events in a simulated environment. In the proposed artificial prediction market, traders who showcase superior predictive abilities prosper, gaining wealth and influence, whereas the less adept face potential bankruptcy and market exit. The inherent dynamics of our market provide a powerful mechanism to identify and differentiate between high- and low-quality comments based on traders' performance, enhancing the reliability of crowd wisdom extracted. In addition, another advantage of the artificial market is that compared to other opinion aggregation methods like the prediction poll proposed by Atanasov et al. [12], our market does not have many hyper-parameters that require additional training.

To evaluate our method, we assess its performance in predicting the stock price movements of constituent stocks in the Shanghai Stock Exchange 50 Index. All artificial human traders in our market are sourced from the stock forum of Eastmoney.com. The designed artificial prediction market aims to leverage the collective wisdom within online communities to predict stock price movements across different time horizons (1 day, 5 days, 15 days, 25 days, and 30 days ahead). Experimental results indicate that the proposed market outperforms comparable benchmark models, effectively distinguishing between high- and low-quality comments.

This study yields three primary contributions. First, to the best of our knowledge, this is the first attempt to construct an artificial prediction market to extract the wisdom of online crowds. We utilize the market mechanism to aggregate the wisdom embedded in online communities for stock price movement prediction, thus contributing a novel approach for opinion mining and stock market prediction research. By creating artificial human traders based on posters' comments, the artificial prediction market mitigates the challenges arising from the abundance of low-quality comments, enhancing the reliability and utility of the aggregated collective wisdom.

Second, we revisit the comparison of the effectiveness between the prediction market and the prediction poll in aggregating opinions. Prior studies have shown that team prediction polls, with parameter tuning and group cooperation, can achieve performance comparable to prediction markets, especially in long-term forecasting tasks [12]. This study indicates that the prediction market can effectively filter out underperforming traders and better identify high-quality comments in long-term, medium-term, and short-term forecasts. Conversely, the prediction poll cannot effectively assign weights to forecasters in short-term forecasts.

Third, the artificial prediction market constructed with posters' comments offers a novel tool to improve the decision-making quality in versatile areas, including but not limited to forecasting geopolitical events, economic policies, and commodity price fluctuations. This research intrinsically aligns with Decision Support Systems principles, which are pivotal in bolstering decision-making processes [13], as it endeavors to optimize the extraction and utilization of collective wisdom from online communities to reinforce decision-making. According to Berg and Rietz [14], a well-structured prediction market can effectively support decision-making processes. Gottschlich and Hinz [1] also advocate the utility of prediction markets in leveraging collective wisdom to improve decision outcomes. Hence, the proposed prediction market provides a novel tool for decision-making support in other prediction scenarios and decision-making processes.

The remainder of this paper is structured as follows: Section 2 reviews related work. Section 3 outlines the problem formulation. In Section 4, we introduce the proposed artificial prediction market, followed by the experimental setup and evaluation criteria in Section 5. Section 6 details the experimental results. Section 7 delves into the efficacy of the proposed market. Finally, we conclude the paper with a summary and directions for future research.

2. Related work

2.1. Sentiment analysis for stock comments

Sentiment analysis, emerging from text-mining and natural language processing, involves detecting and studying opinions, sentiments, emotions, and subjectivities in a text, with the goal of determining the sentiment of the text [15]. Recently, some studies have used sentiment analysis to extract opinions from online stock comments for stock market prediction [16,17]. Sentiment analysis methods are mainly divided into two types [18]: the lexicon approach and the machine learning (including deep learning) approach.

The lexicon approach involves calculating the sum of positive and negative sentiments in a text, where a value greater than zero indicates that the text tends to be positive; otherwise, it is negative [18]. Currently, some studies have begun to focus on the research of lexicon methods in stock market prediction. For example, Wang et al. [19] proposed a novel framework for predicting financial market trends by combining lexicon-based social media sentiment and market technical indicators. Related research includes studies like Oliveira et al. [20] and Deng et al. [21]. One benefit of the lexicon approach is its straightforwardness and independence from labeled texts. However, this method mainly relies on manually constructed lexicons, often leading to insufficient coverage of sentiment words [18].

The machine learning approach mainly involves converting a large number of text features into feature vectors based on sentiment lexicons, and then training a classifier [18]. For the study of sentiment analysis in stock comments, machine learning models are the most widely used [22]. For instance, Jin et al. [23] proposed a deep learning-based stock market prediction model that considers investors' emotional tendency. Compared to lexicon-based approaches, machine learning methods for sentiment analysis, especially of typically short and highly heterogeneous texts, necessitate extensive training corpus and face significant challenges [21]. Additionally, most previous research typically fused the extracted investor sentiment with numerical data, like technical indicators [22], to predict stock market trends. This multi-source data fusion method might overlook nuanced insights from individual comments [24] and compromise data quality, especially during data transformation and dimensionality reduction [25]. Moreover, it could even impair the model's predictive accuracy [26].

2.2. Voting-based methods

Voting-based methods refer to approaches for investor sentiment aggregation, primarily including the Hard Voting approach and the Soft Voting approach. These methods aggregate multiple opinions or decisions, often from a crowd or group, to form a final decision or sentiment. Such voting-based methods have been widely used in stock market prediction [2,4,6]. The Hard Voting method, a commonly used approach, does not consider the quality of comments, treating all investor opinions with equal weight [27]. While its simplicity allows for straightforward interpretations, this method has notable disadvantages. Specifically, the Hard Voting method often overlooks the intrinsic quality of comments, and does not consider the varying expertise levels of posters, making it susceptible to the influence of low-quality comments [4,6].

In contrast to the Hard Voting approach, the Soft Voting method offers a refined approach by incorporating weighted mechanisms, accounting for variability in comment quality or poster expertise. For instance, Chang et al. [6] introduced a novel method, named correlation-based robust dynamic qualities, to model the qualities of investor sentiments, with Hard Voting chosen as one of the benchmark models in its evaluation. Typically, in a Soft Voting model, the individual weights are determined by the poster's expertise level [28], their follower count [29], or a combination of metrics, which may include historical accuracy and posting frequency [12]. For example, Hill and Ready-Campbell [2] proposed a genetic algorithm capable of identifying appropriate vote weights for posters based on their past voting success. Although the Soft Voting method considers comment quality when aggregating investor sentiments, the process of assessing comment quality is typically complex [6,12]. Additionally, some implementations of the Soft Voting method, such as prediction polls, even require training hyper-parameters [12].

2.3. Multi-agent systems and prediction markets

2.3.1. Multi-agent systems in stock markets

Using multi-agent systems in stock market prediction is not new. Luo et al. [30] designed a Multi-Agent System for Stock Trading that supports dynamic information and knowledge exchange among cooperating agents. Kanzari et al. [31] developed an adaptive agent-based model that combines three forecasting behaviors of financial agents: fundamentalist, chartist, and mimetic, and then assessed its ability to predict and explain the dynamics of stock market price formation. In order to handle complex financial scenarios, Huang et al. [32] proposed a multi-agent trading system consisting of multiple agents capable of self-learning and interaction. From existing research, multi-agent systems in stock markets are primarily used to study and analyze the complexity of financial market systems, as well as the interactions between agents themselves and between agents and their environment.

2.3.2. Prediction markets and multi-agent systems

Prediction markets have been used to construct multi-agent systems, also known as artificial prediction markets. The artificial prediction market is a novel method for fusing prediction information from trained classifiers, where the fusion result is the aggregate prediction of the classifiers [33]. In existing artificial prediction markets, traders are primarily composed of machine learning models [33,34] or software trading agents [35,36]. Furthermore, previous research has mainly focused on analyzing the effect of some key factors, such as information-related parameters, on trader behavior [35], as well as studying the impact of key parameters, such as the market liquidity parameter, on the performance of the prediction market [37]. For instance, Jumadinova and Dasgupta [35] constructed an artificial prediction market and studied the effect of various information-related parameters, like arrival rate and reliability of information, on trader behaviors.

Unlike previous multi-agent systems composed of trained machine traders, the traders in the proposed artificial prediction market are constructed based on comments posted by users in online communities, capturing authentic human opinions. Furthermore, our artificial prediction market is mainly used to distill the collective wisdom within online communities, offering a new perspective for opinion mining. Additionally, to the best of our knowledge, this paper represents the first attempt to apply the artificial prediction market for the stock price movement prediction, aiming to test the efficacy of this approach in stock market prediction.

3. Problem formulation

This paper aims to construct an artificial prediction market based on online comments and then utilize the market to predict stock price movements. Aligning with the study by Seong and Nam [38], we predict stock price movements based on five prediction horizons: short-term (1 day and 5 days ahead), medium-term (15 days ahead), and long-term (25 days and 30 days ahead). In each prediction horizon, we construct artificial human traders based on online comments from trading day $t-tw+1$ to trading day t, where tw is the comment time window. These traders then participate in contract trading in our artificial prediction market, aiming to predict the movement of the closing price on trading day $t+h$ relative to the closing price on trading day t, where h denotes the prediction horizon. The predicted target $\hat{y}(t,h)$ is constructed to capture the movement of the closing price $price_t$ within the prediction horizon h, as illustrated in Eq. (1).

$$\hat{y}(t,h) = \begin{cases} 1, & if\ price_{t+h} > price_t \\ 0, & if\ price_{t+h} \leq price_t \end{cases} \tag{1}$$

In trading, artificial human traders will enter the market to buy contract $\omega_{bullish}$, related to the "rise" of stock prices, or contract $\omega_{bearish}$, related to the "fall" of stock prices. The final prices of $\omega_{bullish}$ and $\omega_{bearish}$ represent the probabilistic predictions on stock price movements.

4. Designing artificial prediction market to extract the wisdom of crowds

4.1. The framework of the proposed artificial prediction market

In contrast to traditional artificial prediction markets where traders are primarily algorithm-based, in our artificial prediction market, traders trade based on beliefs extracted from the comments of their corresponding online posters. The proposed framework is illustrated in Fig. 1. It comprises four main components: data collection, belief extraction, market design, and performance evaluation. The initial phase focuses on data collection, encompassing two primary types of data: comment data for constructing artificial human traders and stock market data mainly for market evaluation.

In the second phase, labeled as "Belief Extraction", the objective is to derive posters' beliefs about potential stock price movements directly from their comments. These beliefs, in essence, represent posters' probabilistic predictions about whether a stock will rise or fall. To achieve this, we employed a fine-tuned BERT model and a BiLSTM model, i.e., BERT-BiLSTM.

In "Market Design", our primary focus is on developing a trading

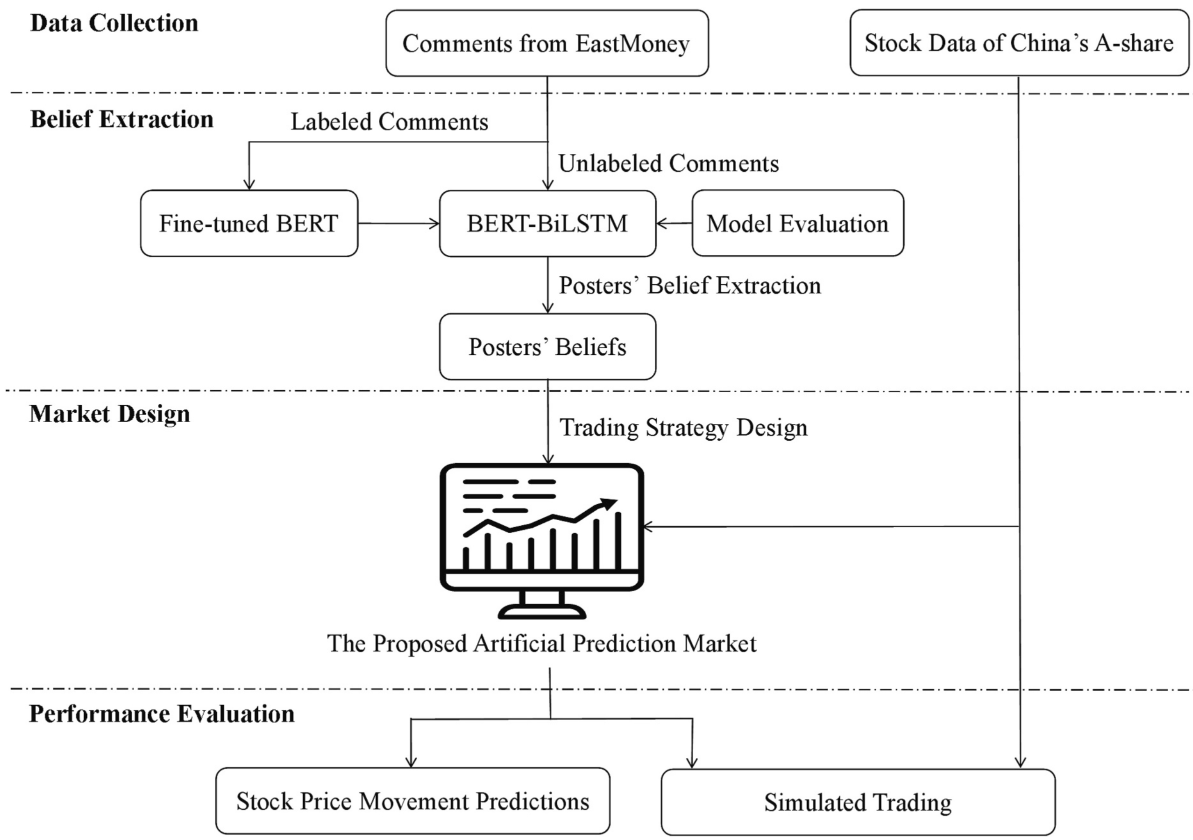


Fig. 1. Framework of the proposed artificial prediction market.

strategy based on economic modeling methods. This strategy integrates traders' beliefs—extracted during the "Belief Extraction" phase—with factors such as their disposable wealth and current market prices. The core idea is to enable traders to buy contracts that are deemed optimal, given their unique beliefs and market environment, ultimately maximizing their potential payoffs.

Finally, this paper evaluates the performance of the proposed artificial prediction market from two dimensions: stock price movement predictions and simulated trading. The final prices in our market effectively integrate the collective wisdom of the traders (or posters) about the stock price movement predictions. Based on this, we initially evaluate the performance of our market regarding predictability. Furthermore, by executing simulated trading based on stock price movement predictions, we further validate the effectiveness of our method in profitability.

4.2. Belief extraction

The process of belief extraction involves the following steps: Comments, truncated to 100 words, are first passed through the BERT model. Following this, the BERT model outputs 768-dimensional feature vectors. These feature vectors are then further processed by a Bidirectional LSTM (BiLSTM) layer. After processing through the BiLSTM, the data is passed to a fully connected layer with 256 hidden units. This layer utilizes a Softmax function to calculate the output probabilities for three categories: positive (bullish), neutral, and negative (bearish). In the process, we utilize the BERT-Base-Chinese model, a pre-trained model provided by Google, which was downloaded from HuggingFace (https://huggingface.co).

While the BiLSTM model does output the probability of neutrality, we do not include neutral comments in our market. As a result, the traders admitted into the artificial prediction market are solely constructed based on bullish and bearish comments. To ensure the sum of traders' beliefs matches the sum of the prices of the two contracts ($\omega_{bullish}$ and $\omega_{bearish}$), we further normalize the traders' bullish and bearish beliefs using the Softmax function. As a result, the belief of an artificial human trader u is calculated as $belief = [prob_u(\omega_{bullish}), prob_u(\omega_{bearish})]$.

4.3. The design of the artificial prediction market

4.3.1. Trading process in the artificial prediction market

Previous studies have shown that the choice of the market mechanism is crucial for the performance of the prediction market [10]. Considering that the logarithmic market scoring rule (LMSR) has high prediction accuracy and good robustness to the setting of parameters [39]. Therefore, we selected LMSR as our market trading mechanism.

In our market, each trader is endowed with a set amount of virtual wealth when they enter the market for the first time. They decide the best contract trading volume based on their beliefs, disposable wealth, and market prices. After the revelation of the actual outcome of the predicted event, the contracts traders hold will be cleared and settled according to the market clearing rules: If the stock's closing price rises, bullish contracts are valued at 1 coin, and bearish ones at 0; if it falls, the values switch. In trading, traders can only buy contracts and cannot sell them. This design fits our two-contract LMSR market, where selling a certain number of contracts and purchasing another contract with the same number would have the same impact on market prices.

In addition, traders can only use their wealth in all markets that predict the same stock and have the same prediction horizon. For example, traders' virtual wealth can only be used in markets that predict the stock price movement 5 days ahead for a given stock, ensuring that the traders' wealth reflects their knowledge and understanding of the stock's movements at that prediction horizon. The trading process is summarized in Algorithm 1.

$price_t(\omega_e)$ can be understood as the consensus probability estimate that traders believe event e will occur at time t. When a trader buys a limited number of contracts $q_t(\omega_e) > 0$ from the market, the cost is [39,40]:

$$C(q_t(\omega_e)) = b^* ln\left(price_t(\omega_e)\left(e^{\frac{q_t(\omega_e)}{b}} - 1\right) + 1\right) \tag{3}$$

Algorithm 1. Trading of a prediction horizon for a target stock in the artificial prediction market.

Input: Comments of the target stock;
 Initial wealth W_0;
 Market liquidity parameter b;
 Bankruptcy prevention parameter β.
Initialize virtual wealth memory $\mathcal{W} = [\,]$.
For $market = market_1$ to $market_M$ **do**:
 For $poster\_com = comment_1$ to $comment_N$ **do**:
  Input $poster\_com$ into the BERT-BiLSTM model to calculate the probability distribution, i.e., $prob_{bnb} = [prob_{bullish}, prob_{neutral}, prob_{bearish}]$.
  If $poster\_com$ is not neutral, i.e., $max\,(prob_{bnb}) \neq prob_{neutral}$:
   Normalizing $prob_{bullish}$ and $prob_{bearish}$ with the Softmax function to get trader u's belief, i.e., $belief = [prob_u(\omega_{bullish}), prob_u(\omega_{bearish})]$.
   Update trader u's budget as:

$$W_u \leftarrow \begin{cases} \text{Initial wealth } W_0, & \text{if trader } u\text{'s ID not in } \mathcal{W}. \\ \text{Update trader } u\text{'s wealth through } \mathcal{W}, & \text{if trader } u\text{'s ID in } \mathcal{W}. \end{cases}$$

   Calculate trader u's optimal trading volume $q^*(\omega_e)$.
   Trader u enters the market and buys the contract ω_e.
 Market clearing.
 Update virtual wealth memory $\mathcal{W}$.
End for.
Output: Final market price vector $\boldsymbol{P} = \{[price_T(\omega_{bullish}), price_T(\omega_{bearish})]_m | m = 1,2,\cdots M\}$.

Note: M denotes the number of trading days; N represents the number of comments; T refers to the time of the last trade in a market.

4.3.2. *Trading strategy in the artificial prediction market*

This section will derive the optimal contract trading volume $q^*(\omega_e)$ as shown in Algorithm 1. Our market is primarily used for predicting stock price movements: *bullish* and *bearish*. Therefore, the outcome space is $\Omega = \{\omega_{bullish}, \omega_{bearish}\}$, where $\omega_{bullish}$ is the bullish contract, and $\omega_{bearish}$ represents the bearish contract. Hence, for a given time t, the prices for contracts $\omega_{bullish}$ and $\omega_{bearish}$ are:

$$price_t(\omega_{bullish}) = \frac{e^{\frac{q_t(\omega_{bullish})}{b}}}{\sum_{\tau\in\Omega} e^{\frac{q_t(\tau)}{b}}}; price_t(\omega_{bearish}) = \frac{e^{\frac{q_t(\omega_{bearish})}{b}}}{\sum_{\tau\in\Omega} e^{\frac{q_t(\tau)}{b}}} \tag{2}$$

where $q_t(\omega_e)$ ($e =$ *bullish* or *bearish*) is the total traded amount of contract ω_e at time t, b denotes the market liquidity parameter, and

Assuming that the goal of each trader is to maximize their expected return, trader u's objective function when buying contract ω_e can be expressed as follows:

$$\max_{q_t(\omega_e)} prob_u(\omega_e)(q_t(\omega_e) - C(q(\omega_e))) + (1 - prob_u(\omega_e))(-C(q_t(\omega_e)))$$

$$s.t. C(q_t(\omega_e)) \leq \beta W_{u,t} \text{ and } q_t(\omega_e) \geq 0 \tag{4}$$

where $prob_u(\omega_e)$ represents the probability that trader u believes the event e will occur; $W_{u,t}$ represents the total wealth owned by trader u at time t; and β is referred to as the bankruptcy prevention parameter, which helps prevent trader u from going bankrupt early. By solving Eq. (4), we can get trader u's optimal contract trading volume $q_t^*(\omega_e)$:

$$q_t^*(\omega_e) = \begin{cases} 0, & price_t(\omega_e) > prob_u(\omega_e) \\ b^* ln\dfrac{prob_u(\omega_e)(1 - price_t(\omega_e))}{(1 - prob_u(\omega_e))price_t(\omega_e)}, & g(prob_u(\omega_e)) \le price_t(\omega_e) \le prob_u(\omega_e) \\ b^* ln\left(\dfrac{1}{price_t(\omega_e)}\left(e^{\frac{\beta W_{u,t}}{b}} - 1\right) + 1\right), & price_t(\omega_e) < g(prob_u(\omega_e)) \end{cases} \tag{5}$$

where $g(prob_u(\omega_e)) = 1 - (1 - prob_u(\omega_e))e^{\frac{\beta W_{u,t}}{b}}$. In Eq. (5), when $price_t(\omega_e) > prob_u(\omega_e)$, trader u will not buy the corresponding contract. When $g(prob_u(\omega_e)) \le price_t(\omega_e) \le prob_u(\omega_e)$, trader u will buy the optimal contract volume and still have disposable wealth left, i.e., $\beta W_{u,t} - C(q_t^*(\omega_e))$. Furthermore, when $price_t(\omega_e) < g(prob_u(\omega_e))$, trader u will use all disposable wealth $\beta W_{u,t}$ to buy the corresponding contract.

5. Experimental setup and evaluation

5.1. Data set

The price movements of the constituent stocks of the Shanghai Stock Exchange 50 (SSE50) Index are used as forecasting tasks for our artificial prediction market. The SSE 50 Index consists of 50 stocks from China's A-shares, listed on the Shanghai Stock Exchange, representing prominent and influential enterprises in the Shanghai stock market. Among these 50 firms, we excluded 11 firms listed after 2012, as these 11 firms had an average of fewer than five comments per day. Consequently, we conducted experiments on the remaining 39 firms.

The Scrapy package in Python was used to collect comments from the "GuBa" forum on EastMoney (https://www.eastmoney.com), which is China's largest stock message board. A total of 1,874,629 comments related to 39 target stocks were collected, all posted by users between January 1, 2016, and April 29, 2022. Additionally, to conduct market clearing and evaluate the market performance, we collected adjusted closing prices for these 39 firms. We then divided the comment and stock data into training and testing sets. Specifically, the data from January 1, 2016, to December 31, 2019, was used as the training set for optimizing the parameters of each method, while the data from January 1, 2020, to April 29, 2022, served as the testing set to evaluate the performance of the proposed artificial prediction market as well as other benchmark models.

To train the BERT-BiLSTM model, we also constructed a dataset of 12,000 manually labeled comments. These 12,000 comments were randomly selected from the training set. To label the comments, we recruited six senior undergraduates majoring in finance to label them on the easyDL platform (https://ai.baidu.com). The BERT-BiLSTM model is fine-tuned by using those 12,000 manually labeled comments, and it performs well, achieving an average accuracy of 94.56% based on 10-fold cross-validation evaluation.

5.2. Implementation details of the proposed artificial prediction market

The performance of the artificial prediction market, like machine learning algorithms, is influenced by various factors such as the number of traders, liquidity parameter b, and traders' initial wealth W_0 [41]. Among these factors, b and W_0 are critical. Additionally, the comment time window tw, representing a time interval during which traders constructed based on comments from this interval will trade in the same market, is also important in our market.

To optimize these crucial parameters, we employed a grid search method on the training set to explore a range of potential values: {100, 200, 300, …, 1000} for b; {10, 20, 30, …, 150} for W_0; and {5, 10, 15, 25, 30} for tw. We finally selected the parameters that made our artificial prediction market produce the lowest Brier score on the training set.

The Brier score measures the calibration of prediction outputs [42], calculated as the mean square difference between subjective probabilities and actual outcomes [43].

$$Briescore = \frac{1}{N}\sum_{n=1}^{N}(\hat{y}_n - y_n)^2 \tag{6}$$

where N denotes the number of trading days being predicted, y_n represents the actual results, and $\hat{y}_n$ represents the predicted results (probability). The Brier score was chosen as the criterion for selecting parameters because it is more sensitive to the level of the estimated probabilities [44]. Furthermore, previous literature on prediction markets frequently adopts the Brier score as a performance metric [12]. Finally, we obtain the optimal parameters for each prediction horizon based on minimizing the Brier score, as presented in Table 1.

Furthermore, we also visualized the relationship between parameters (b and W_0) and the market performance on the training set in Fig. 2. We can derive two design principles based on the results from Table 1 and Fig. 2. First, regardless of whether it is for short-term, medium-term, or long-term forecasts, it is necessary to set a larger liquidity parameter b as much as possible. The reason for setting a larger b in the short-term is mainly to prevent market price manipulation by influential individual traders, while in the medium-term and long-term, it is to avoid price collapse, i.e., when $q_t(\omega_e) \to +\infty$, $e^{\frac{q_t(\omega_e)}{b}} \to +\infty$. Second, regarding the settings for W_0 and tw, we find that traders need to be endowed with more W_0 as tw increases. Providing traders with more W_0 under a larger tw helps them to fully utilize their predictive ability.

5.3. Benchmark algorithms

To evaluate our market, we primarily chose five benchmark models for comparative analysis: Prediction Poll, Hard Voting, Soft Voting by Comment Clicks, and two BERT-NN models.

5.3.1. Prediction Poll (Pred-Poll)

The Pred-Poll proposed by Atanasov et al. [12] is a Soft Voting model aggregating individual predictions (or artificial human traders) into a single forecast:

$$\overline{p}_{\ell,e,l} = \frac{\widetilde{p}^{a}{}_{\ell,e,l}}{\widetilde{p}^{a}{}_{\ell,e,l} + \left(1 - \widetilde{p}_{\ell,e,l}\right)^{a}} \tag{7}$$

Table 1
The optimal parameters for the artificial prediction market.

Parameters	1 day ahead	5 days ahead	15 days ahead	25 days ahead	30 days ahead
b	1000.0	900.0	800.0	1000.0	1000.0
W_0	10	60	90	130	140
tw	5 trading days	10 trading days	25 trading days	30 trading days	30 trading days
AveBriescore	0.241591	0.218063	0.205070	0.168071	0.160704

Note: *AveBriescore* represents the average Brier score on the training set of the 39 target stocks.

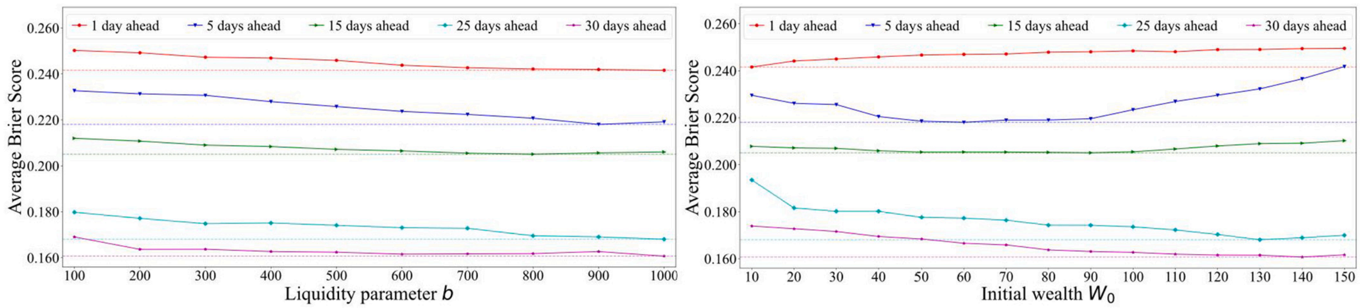


Fig. 2. Liquidity parameter b, initial wealth W_0 and the market performance.

Table 2
Parameter selection ranges for training the Pred-Poll model.

Parameters	Range
learning rate	$1 \times 10^{-5} \sim 1 \times 10^{-2}$
optimizer	$[Adam, SGD, RMSprop]$
net_alpha	$1 \times 10^{-3} \sim 1 \times 10^{3}$
number of epochs	$[10, 20, 30, 50, 80, 100]$
batch size	$256 \sim 1024$
l1_ratio	$[0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9]$

where a is the recalibration parameter, ℓ represents the time at the end of belief aggregation, e = *bullish* or *bearish*, l indicates the target stock. And $\widetilde{p}_{\ell,e,l}$ signifies the weighted probability aggregate across all forecasters:

$$\widetilde{p}_{\ell,e,l} = \frac{\sum_{t=1}^{tw} \sum_{u=1}^{N_t} d_{t,u,l} \times \omega_{t,u,l} \times p_{t,u,e,l}}{\left(\sum_{t=1}^{tw} \sum_{u=1}^{N_t} d_{t,u,l} \times \omega_{t,u,l}\right)} \tag{8}$$

where N_t represents the number of forecasters, tw is the comment time window, $d_{t,u,l}$ is the temporal decay parameter, $p_{t,u,e,l}$ indicates the belief of forecaster u. The weight of a forecast, $\omega_{t,u,l}$, depends on forecaster u's prior accuracy $s_{t,u}$ and belief updating frequency $f_{t,u,l}$:

$$\omega_{t,u,l} = {s_{t,u}}^{\alpha} \times {f_{t,u,l}}^{\beta} \tag{9}$$

where $s_{t,u}$ was raised to the exponent α, and $f_{t,u,l}$ was raised to the exponent β. The prior accuracy $s_{t,u}$ was based on forecaster u's average historical Brier score on all the comments posted by u before time t, including those in forums of other stocks besides the 39 target stocks. We collected 22,231,532 comments from non-target stock forums by tracking the IDs of posters who previously commented on the 39 target stocks. The belief updating frequency $f_{t,u,l}$ refers to the posting frequency of the forecaster u before the time t within a comment time window.

Regarding the temporal decay parameter $d_{t,u,l}$, we set $d_{t,u,l}$ to 1 for recent forecasts, and 0 otherwise. Additionally, we optimized α and β on the training set under various combinations of proportions of recent forecasts and recalibration parameters. The set of proportions of recent forecasts is $\{0.1, 0.2 \ldots 1.0\}$, and the recalibration parameter set is $\{0.1, 0.2, 0.3 \ldots 10.0\}$. α and β were optimized under the Optuna framework (https://optuna.org). To find the optimal α and β, we conducted 50 trials, and each trial underwent time series 10-fold cross-validation, with the validation set being set to 2 months. Table 2 presents the parameter selection ranges for training the Pred-Poll model. Elastic Net was employed at each training step to mitigate overfitting, following the approach proposed by Zou and Hastie [45] and used by Atanasov et al. [12].

In training the Pred-Poll model, we set the comment time window to 30, as our findings revealed that increasing the window size did not further improve performance. Table 3 presents the optimal parameters obtained for the Pred-Poll model. Additionally, Fig. 3 illustrates the impact of the proportions of recent forecasts and recalibration parameters on the aggregate performance of the Pred-Poll model in the validation set.

5.3.2. *Hard Voting (Hard-Voting)*

The Hard-Voting model adopts an equal-weighted summation technique to aggregate all investor sentiments. In this approach, each comment from day $t - tw + 1$ to day t is treated as an individual forecaster, and the comments related to the target stock s within this period are aggregated using a scoring function, as expressed by Eq. (10).

$$hv(s, e) = \frac{1}{num\_coment_s} \sum_{c \in coment_s} \mathbb{I}\{a_c = e\} \tag{10}$$

where $e \in \{bullish, bearish\}$; a_c denotes comment c's opinion, i.e., $a_c \in \{bullish, bearish\}$; $coment_s$ refers to the comments related to the stock s posted by all posters during the period from $t - tw + 1$ to t; the standardization factor $1/num\_coment_s$ can ensure $\sum_e hv(s, e) = 1$; $\mathbb{I}$ indicates the indicator function.

5.3.3. *Soft Voting by Comment Clicks (Soft-VCC)*

Unlike the Hard-Voting model, the Soft-VCC model aggregates all investor sentiments by taking the clicks of each comment as its weight.

$$sv(s, e) = \sum_{c \in coment_s} \widehat{q}(c) \mathbb{I}\{a_c = e\} \tag{11}$$

where $\widehat{q}(c)$ represents the quality or influence of the comment c.

In previous studies, some researchers have aggregated investor sentiments based on posters' followers [29]. While follower count might

Table 3
The optimal parameters for the Pred-Poll model.

Parameters	1 day ahead	5 days ahead	15 days ahead	25 days ahead	30 days ahead
Proportions of recent forecasts	0.4	0.9	0.7	1.0	1.0
α	−0.603790	−0.656758	−1.878723	−3.933201	−3.584083
β	−0.112103	−0.228610	0.113287	1.222056	1.285019
a	1.5	1.6	5.8	7.5	7.8
AveBriescore	0.236292	0.235589	0.201747	0.200320	0.197958

Note: *AveBriescore* represents the average Brier score of the Pred-Poll model on the validation set of the 39 target stocks, calculated while maintaining all other parameters at their optimized levels.

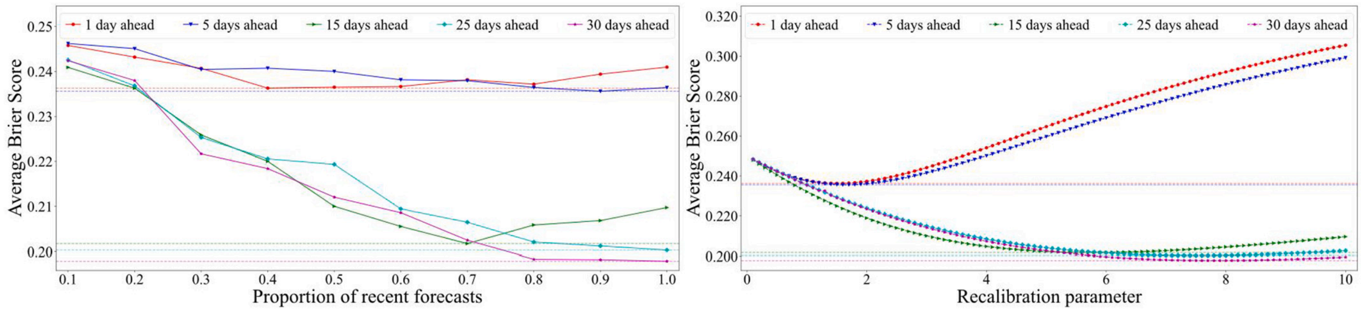


Fig. 3. Performance of the Pred-Poll: proportions of recent forecasts vs recalibration parameters.

Table 4
Parameter selection ranges of the BERT-NN2 and BERT-NN3 models.

Parameters	Range
number of hidden layers	$256 \sim 1024$
learning rate	$1 \times 10^{-5} \sim 1 \times 10^{-2}$
activation function	$[ReLU, Sigmoid, Tanh]$
optimizer	$[Adam, SGD, RMSprop]$
dropout rate	$[0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9]$
number of epochs	$[300, 400, 500, 600]$
batch size	$256 \sim 1024$

suggest a poster's influence within a community, it is worth mentioning that a comment can influence even those who are not followers. Additionally, not every follower will browse the poster's comments. Considering these factors, this paper suggests employing the number of clicks on each comment as a weight, denoted as $\widehat{q}(c) = \frac{click_c}{\sum_{c \in coments_s} click_c}$. Here, $click_c$ is the number of clicks on comment c, including both followers and non-followers. This approach allows us to capture the influence of a comment across the entire community. Therefore, when $\widehat{q}(c)$ is incorporated into Eq. (11), we obtain the following result:

$$sv(s,e) = \sum_{c \in coments_s} \frac{click_c}{\sum_{c \in coments_s} click_c} \mathbb{I}\{a_c = e\} \tag{12}$$

5.3.4. BERT-NN (2–3)

The last two benchmarks are models named BERT-NN2 and BERT-NN3. In the former, we added two fully connected layers (NN2) after the BERT layer, while in the latter, three fully connected layers (NN3) were incorporated. Inspired by the research of Mai et al. [46], we employed the fine-tuned BERT model to extract features from each comment within the same comment time window. We then averaged these features, resulting in an average sentence embedding fed into the two models for binary classification: bullish or bearish.

Since the BERT model has already been fine-tuned, only the fully connected layers require training, with the BERT model solely responsible for extracting features from comments. Therefore, we trained only the NN2 and NN3 models under the Optuna framework through 50 trials, each undergoing time series 10-fold cross-validation, and the validation set was set to 2 months. The parameter selection ranges for training the two models are presented in Table 4.

For comparability, the comment time windows for Hard-Voting, Soft-Voting, BERT-NN2, and BERT-NN3 in predicting stock price movements for 1 day, 5 days, 15 days, 25 days, and 30 days ahead are also set to 5, 10, 25, 30, and 30 trading days, respectively.

5.4. Evaluation criteria

The evaluation of the proposed artificial prediction market revolves around two essential dimensions: predictability (stock price movement predictions) and profitability (simulated trading). First, we compared our market's predictability against other benchmark models using five metrics: accuracy, precision, recall, F-1 score, and Brier score.

Second, to test the profitability of our market, we implemented h independent portfolios in a prediction scenario with a prediction horizon of h. The initial trading days for the h portfolios are respectively t_0, $t_0 + 1$, …, and $t_0 + h - 1$, with each portfolio being rebalanced once after h trading days. During each rebalancing, we initially sold all the stocks predicted to be bearish. Then, guided by the optimal asset allocation determined by the mean-variance (MV) model [47], we utilized the portfolio's total value, including the account balance and the market value of all invested assets, to readjust the holdings in existing bullish stocks. The existing bullish stocks included both newly identified bullish stocks and those determined in the previous period.

Specifically, based on the Markowitz portfolio theory [48], we determined the weights for optimal asset allocation during portfolio rebalancing using the closing prices of the most recent 252 trading days. After obtaining the time series of the portfolio values of the h portfolios, the average portfolio value at each time point was calculated. From the calculated average portfolio value series, five metrics were determined: annual return (AnR), cumulative return (CuR), Sharpe ratio (ShR), stability (Stab), and maximum drawdown (MDD). For this purpose, we implemented 50,000 rounds of Monte Carlo simulation. In each of the 50,000 iterations, a random weight vector was generated, and the corresponding portfolio returns and variances were computed. We selected the weights with the highest Sharpe ratio as our final asset allocation weights. In trading, returns were computed using Eq. (13).

$$RR_i = \left(b_{t+h} + \boldsymbol{p}_{t+h}^T \boldsymbol{h}_{t+h}\right) - \left(b_t + \boldsymbol{p}_t^T \boldsymbol{h}_t\right) - c_t \tag{13}$$

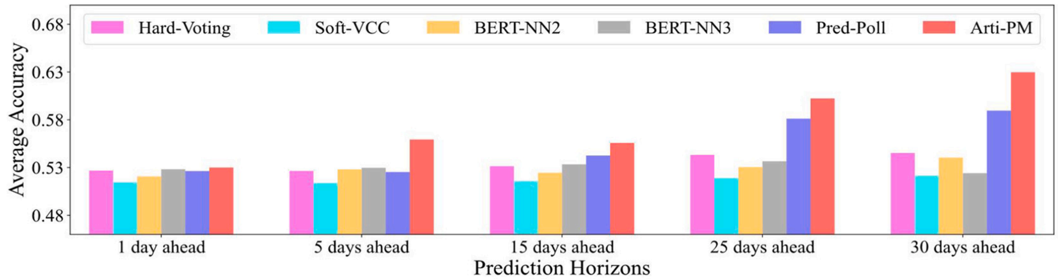
where i represents the proposed artificial prediction market or a benchmark model; b_t denotes the account balance; $\boldsymbol{p}_t$ indicates the price vector of all stocks; $\boldsymbol{h}_t$ signifies the vector of the number of stocks held; c_t stands for the total trading cost; and h is the prediction horizon.

6. Results

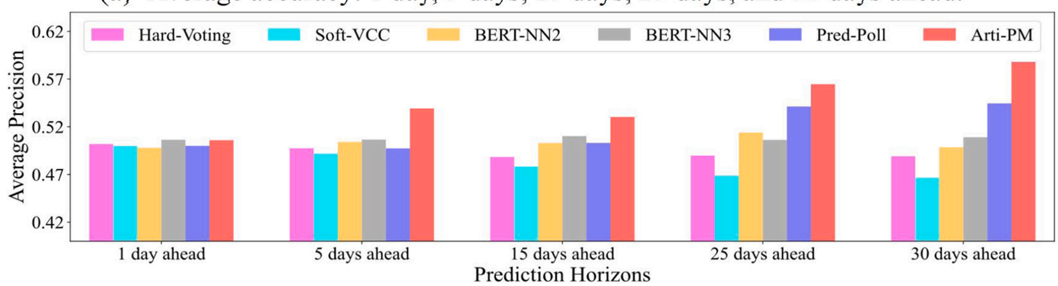
6.1. Stock price movement prediction

This section presents the performance of the proposed artificial prediction market (Arti-PM) in predicting stock price movements, which results from a single run test on the test set, as illustrated in Fig. 4 and Table 5. Due to space limitations, Table 5 only shows the accuracy of each method. As seen from Table 5, whether in short-term, medium-term, or long-term forecasting, the proposed Arti-PM consistently achieves higher accuracy than other benchmark models, significantly outperforming these benchmarks in many instances.

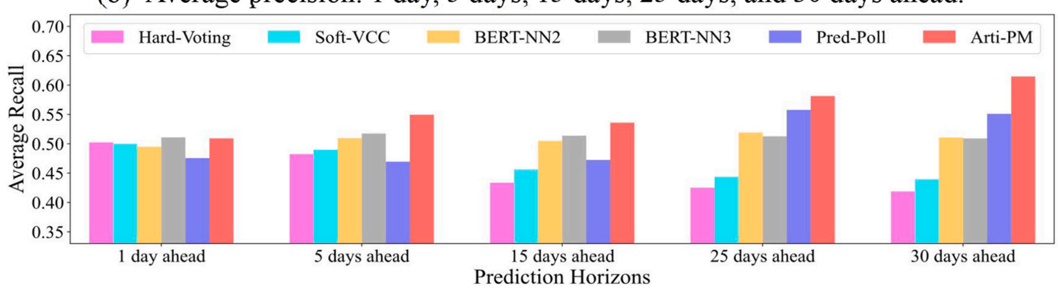
We have drawn two notable conclusions by comparing the performance of each method in stock price movement predictions. First, the Arti-PM demonstrates favorable performance across all metrics, including accuracy, precision, recall, F-1 score, and Brier score. Notably, it holds a pronounced advantage in medium to long-term forecasting. Second, although the Pred-Poll's performance is not as impressive as that of the Arti-PM, it exceeds all benchmarks in medium and long-term



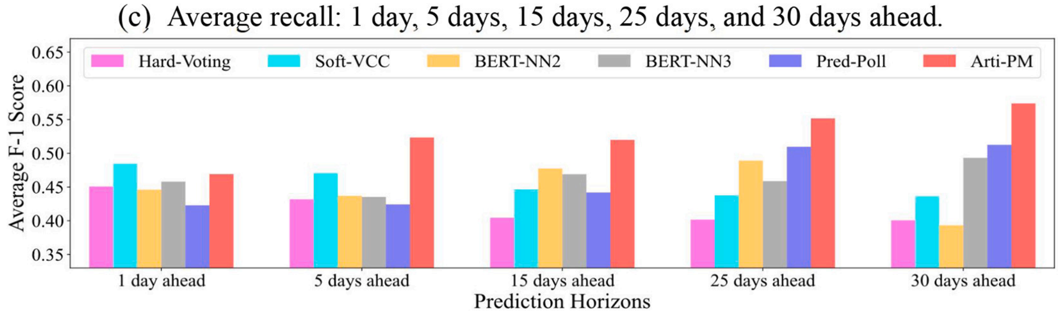
(a) Average accuracy: 1 day, 5 days, 15 days, 25 days, and 30 days ahead.



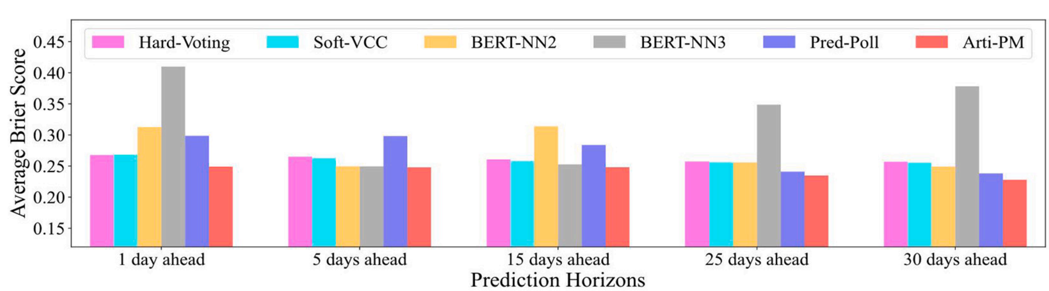
(b) Average precision: 1 day, 5 days, 15 days, 25 days, and 30 days ahead.



(c) Average recall: 1 day, 5 days, 15 days, 25 days, and 30 days ahead.



(d) Average F-1 score: 1 day, 5 days, 15 days, 25 days, and 30 days ahead.



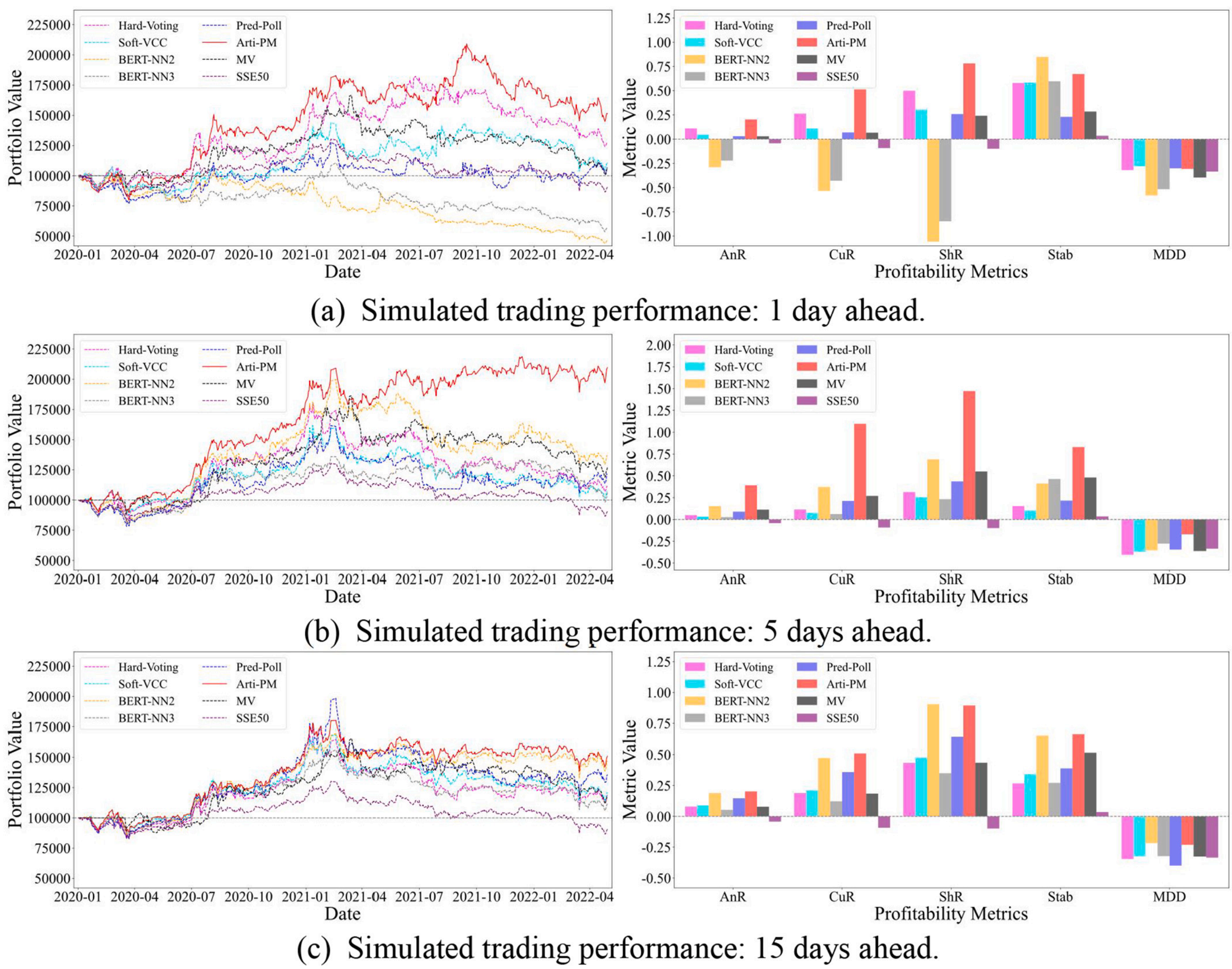
(e) Average Brier score: 1 day, 5 days, 15 days, 25 days, and 30 days ahead.

Fig. 4. Performance comparison of stock price movement prediction.

Table 5
The Wilcoxon tests of accuracy metric.

Prediction Horizons	Hard-Voting	Soft-VCC	Pred-Poll	BERT-NN2	BERT-NN3	Arti-PM
1 day ahead	0.526844 (0.402395)	0.514567 (0.000552)***	0.526331 (0.548375)	0.520705 (0.015750)*	0.528174 (0.494083)	**0.530051**
5 days ahead	0.526472 (8.34e-07)***	0.513838 (2.70e-07)***	0.525318 (1.94e-06)***	0.528074 (1.37e-05)***	0.529751 (9.05e-05)***	**0.559328**
15 days ahead	0.531462 (0.005848)**	0.515797 (3.10e-05)***	0.542628 (0.288857)	0.524518 (0.000423)***	0.533377 (0.004923)**	**0.555787**
25 days ahead	0.543287 (3.14e-06)***	0.518956 (3.78e-07)***	0.581162 (0.017016)	0.530410 (2.71e-07)***	0.536585 (3.51e-07)***	**0.602346**
30 days ahead	0.545238 (1.60e-07)***	0.521495 (9.05e-08)***	0.589585 (0.000194)***	0.540392 (6.90e-08)***	0.524092 (5.25e-08)***	**0.629795**

Note: Values in parentheses are p-statistics between the proposed Arti-PM and other benchmark models. ***$p < 0.001$, **$p < 0.01$, *$p < 0.05$.



(a) Simulated trading performance: 1 day ahead.

(b) Simulated trading performance: 5 days ahead.

(c) Simulated trading performance: 15 days ahead.

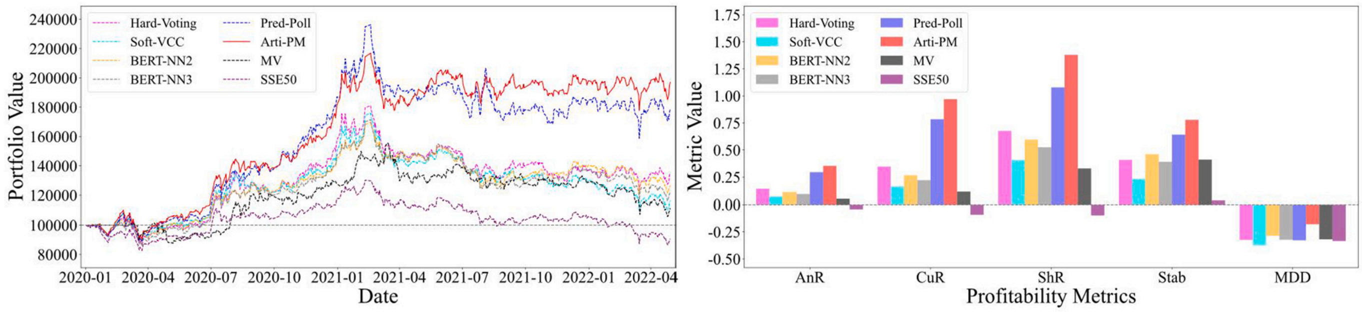
Fig. 5. Simulated trading performance: 1 day, 5 days, 15 days, 25 days, and 30 days ahead.

forecasting. However, its effectiveness is limited in short-term forecasting.

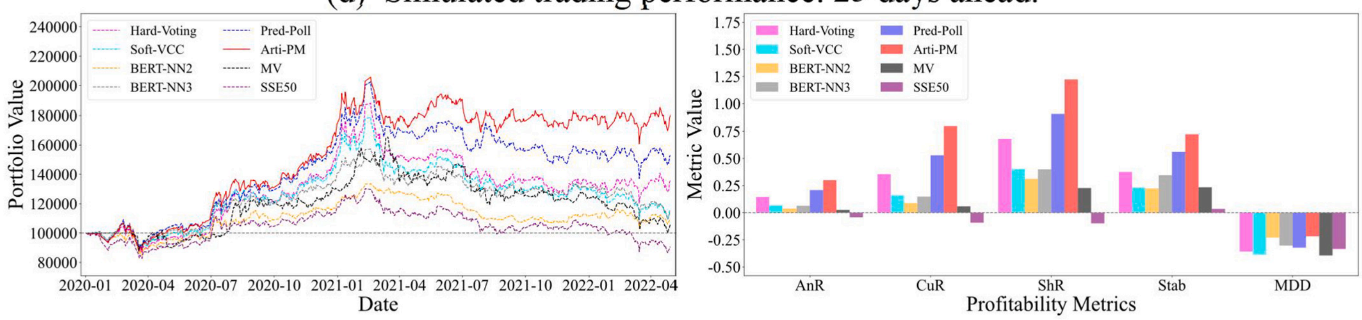
6.2. Results of simulated trading

During the simulated trading, the initial assets of each independent portfolio are set at 100,000. A transaction fee of 0.001 is charged when buying stocks, and a transaction fee of 0.002 is charged when selling stocks. Additionally, in the profitability comparison analysis, we added two extra benchmarks: SSE50 and MV. The SSE50 represents the buy-and-hold strategy for the SSE50 Index. MV refers to a simple MV-optimized portfolio, which is based on all 39 stocks and rebalanced at the same frequency as other strategies, but it ignores all other signals. The results of the simulated trading, derived from a single run test on the test set, are presented in Fig. 5 and Table 6. Table 6 only lists the Sharpe ratio of each method for different prediction horizons.

As shown in Fig. 5, compared to other benchmark models, our Arti-PM exhibits a significant overall advantage. For example, in the portfolios constructed based on the results of predicting stock price movements 1 day ahead, our market achieved the highest Sharpe ratio of 0.781081. Moreover, the Sharpe ratios of the portfolios constructed based on the results of the Arti-PM's stock price movement predictions



(d) Simulated trading performance: 25 days ahead.



(e) Simulated trading performance: 30 days ahead.

Fig. 5. (*continued*).

Table 6
The Sharpe ratios in portfolios.

Prediction Horizons	Hard-Voting	Soft-VCC	BERT-NN2	BERT-NN3	Pred-Poll	Arti-PM	MV	SSE50
1 day ahead	0.500508	0.305491	−1.056762	−0.847170	0.258889	**0.781081**	0.241810	−0.098895
5 days ahead	0.315136	0.254991	0.688594	0.232418	0.436198	**1.471519**	0.549653	−0.098895
15 days ahead	0.434099	0.477057	**0.904156**	0.350256	0.645925	0.894171	0.434905	−0.098895
25 days ahead	0.676628	0.405881	0.596466	0.525119	1.078313	**1.379516**	0.329616	−0.098895
30 days ahead	0.680956	0.402519	0.312156	0.399898	0.911873	**1.222742**	0.227665	−0.098895

Note: The investment on the SSE50 Index employed the buy-and-hold strategy, without any portfolio rebalancing.

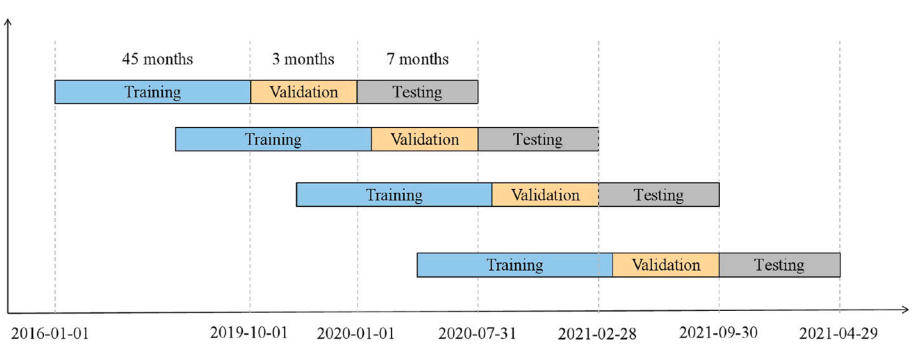


Fig. 6. Illustration of rolling training, validation and testing.

for 5 days, 15 days, 25 days, and 30 days ahead are 1.471519, 0.894171, 1.379516, and 1.222742, respectively, surpassing those achieved by other benchmark models. However, in the portfolios constructed based on the results of predicting stock price movements 1 day ahead, BERT-NN2 achieved the highest Sharpe ratio of 0.904156.

6.3. Rolling window testing

To further evaluate the robustness of the Arti-PM, we conducted tests using the rolling window approach with a seven-month window size (as depicted in Fig. 6). Tables 7 and 8 present each method's accuracy and Sharpe ratio on the test set under the rolling window approach, where the prediction horizon is set to 5. The implementation details are consistent with those in a single run test, including transaction fees and initial assets for each independent portfolio. The results indicate that the proposed Arti-PM outperforms other benchmark models in terms of predictability and profitability.

7. Why does the artificial prediction market perform well?

The superior performance of the Arti-PM can be attributed to traders' wealth, serving as an effective measure of their abilities and dedication

Table 7
Wilcoxon tests for accuracy in each rolling window.

Rolling Windows	Hard-Voting	Soft-VCC	BERT-NN2	BERT-NN3	Pred-Poll	Arti-PM
2020/01/01–2020/07/31	0.513951 (1.42e-06)***	0.516603 (1.21e-05)***	0.52890 (0.000174)***	0.518464 (0.000301)***	0.509205 (2.94e-06)***	**0.578905**
2020/08/01–2021/02/28	0.516341 (5.33e-05)***	0.507946 (4.33e-05)***	0.524564 (0.001463)**	0.518149 (0.076343)	0.516731 (0.000362)***	**0.562538**
2021/03/01–2021/09/30	0.528946 (0.000424)***	0.511292 (2.30e-07)***	0.535951 (0.000424)***	0.545021 (0.031626)*	0.525290 (0.000194)***	**0.567877**
2021/10/01–2022/04/29	0.546579 (0.003020)**	0.519954 (5.01e-05)***	0.537144 (0.001610)**	0.553933 (0.041597)*	0.545564 (0.019036)*	**0.577674**

Note: Values in parentheses are p-statistics between Arti-PM and benchmarks. ***$p < 0.001$, **$p < 0.01$, *$p < 0.05$.

Table 8
Sharpe ratios in each rolling window.

Rolling Windows	Hard-Voting	Soft-VCC	BERT-NN2	BERT-NN3	Pred-Poll	Arti-PM
2020/01/01–2020/07/31	0.534321	0.595236	0.767490	0.690070	0.375089	**1.202402**
2020/08/01–2021/02/28	0.693203	0.527858	1.117071	0.380953	0.610359	**1.863727**
2021/03/01–2021/09/30	−0.927903	−1.553756	−0.444887	0.310428	−1.412481	**0.692871**
2021/10/01–2022/04/29	−1.053969	−0.901372	−0.340582	−1.206099	0.489892	**0.971257**

Table 9
OLS regression results on the effects of α_{pm} and β_{pm} on $wealth_{t,u,l}$.

Variables	1 day ahead	5 days ahead	15 days ahead	25 days ahead	30 days ahead
α_{pm}	−1.666918*** (0.001005)	−2.883770*** (0.000552)	−3.137665*** (0.000615)	−3.370750*** (0.000658)	−3.665375*** (0.000712)
β_{pm}	0.150675*** (0.003635)	0.165907*** (0.001599)	0.156420*** (0.001817)	0.156320*** (0.001920)	0.161324*** (0.002065)
Adj.R-squared	0.946285	0.938893	0.936960	0.935894	0.936135

Note: Robust standard errors are in parentheses. ***$p < 0.01$, **$p < 0.05$, *$p < 0.1$.

to updating their beliefs. Because the market's mechanism inherently allows for the selection of outstanding traders through traders' wealth. The Pred-Poll model considers individuals' prior accuracy and belief updating frequency when determining weights [12]. Consequently, we will discuss both the wealth of traders within the Arti-PM and the weights of forecasters in the Pred-Poll model. The traders in the Arti-PM and the forecasters in the Pred-Poll model refer to the same group of participants.

The Pred-Poll model calculates each forecaster's weight as $\omega_{t,u,l} = {s_{t,u}}^{\alpha} \times {f_{t,u,l}}^{\beta}$, which is quite similar to the Cobb-Douglas production function and provides an intuitive interpretation of the weight. By taking logarithms on both sides, we get:

$$ln\omega_{t,u,l} = \alpha ln s_{t,u} + \beta ln f_{t,u,l} \tag{14}$$

Eq. (14) is a log-log linear regression without an intercept term. Table 3 shows that for predictions 1 day ahead, β equals −0.112103, while for predictions 5 days ahead, β equals −0.228610. Negative β values contradict the expected outcomes because, in theory, as the belief updating frequency increases, the information content of forecasters should also increase, and their weights should increase accordingly. This suggests that β should be greater than 0. Therefore, the weight formula in the Pred-Poll model is not particularly suitable for short-term predictions. In Atanasov et al. [12], the Pred-Poll model was compared with the prediction market in long-term rather than short-term forecasting tasks.

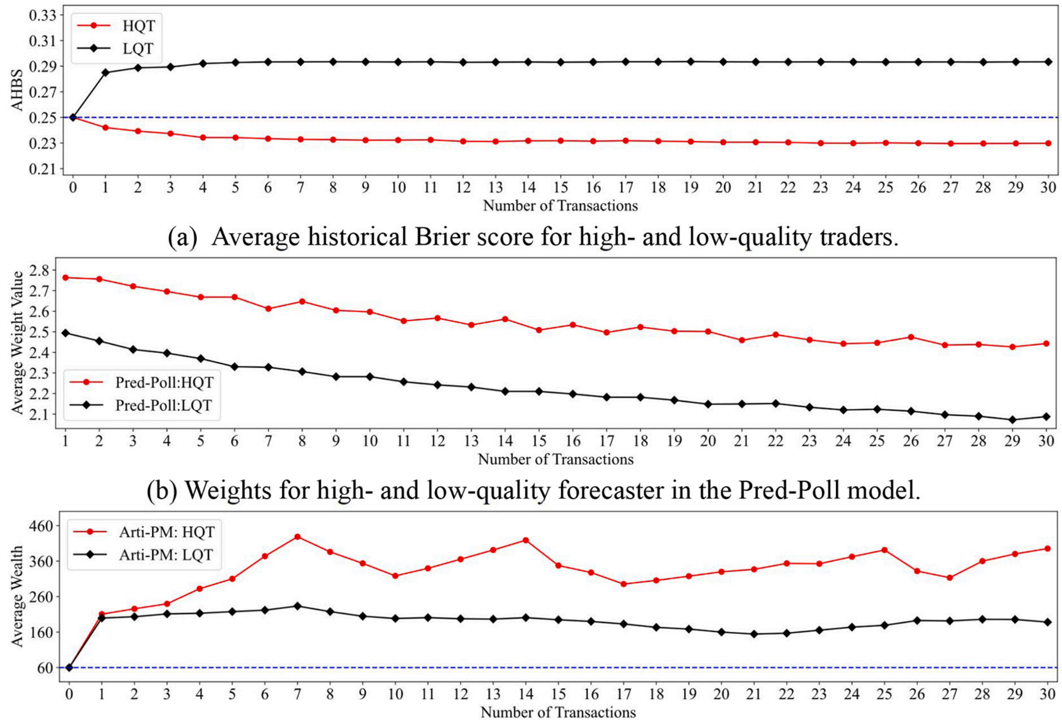
In our Arti-PM, traders' wealth primarily depends on their abilities (prior accuracy) and their efforts in updating beliefs. Using wealth as a proxy for individual influence or weight, we carried out a comparable log-log linear regression without an intercept term.

$$lnwealth_{t,u,l} = \alpha_{pm} ln s_{t,u} + \beta_{pm} ln f_{t,u,l} \tag{15}$$

where $wealth_{t,u,l}$ denotes trader u's wealth at the end of each market. Table 9 presents the regression results for Eq. (15), showing a negative correlation between traders' wealth and their prior accuracy (average historical Brier score), aligning with findings from the Pred-Poll model. Moreover, there is a consistent positive correlation between the belief updating frequency $f_{t,u,l}$ and trader's wealth $wealth_{t,u,l}$ in the Arti-PM, irrespective of the prediction horizon. This observation aligns with the theoretical expectations, indicating that traders can increase wealth by actively seeking information and updating their beliefs. Additionally, except for predictions 25 days ahead, the Arti-PM is more sensitive to traders' prior accuracy than the Pred-Poll model.

To evaluate our market's ability to distinguish between high- and low-quality comments during opinion aggregation, we examined traders (or forecasters) with a total trading frequency exceeding 30 trades. These traders were categorized into high-quality traders (HQT) with a historical average Brier score under 0.25 and low-quality traders (LQT) with scores of 0.25 or more, as a predicted probability of 0.5 corresponds to a Brier score of 0.25 [43]. We visualized these traders' weight in the Pred-Poll and their wealth in the Arti-PM. We only visualize the results from stock price movement predictions 5 days ahead, as shown in Fig. 7.

Fig. 7(a) displays traders' average historical Brier score (AHBS), indicating that the prediction accuracy of high-quality traders gradually increases while decreasing for low-quality traders. Fig. 7(b) reveals that the Pred-Poll model gives more weight to high-quality forecasters than low-quality ones. Nevertheless, with increasing transactions, weights for both high-quality and low-quality forecasters tend to decline. This decline contradicts our expectation that weights for high-quality forecasters should rise, while those for low-quality forecasters should fall as transactions increase. Thus, the Pred-Poll model's weight design appears unsuitable for short-term forecasting. Contrastingly, Fig. 7(c) illustrates that as transactions increase, the wealth of high-quality traders exhibits an upward trend, consistently higher than low-quality traders, with the gap between them increasingly widening. This trend indicates that the



(a) Average historical Brier score for high- and low-quality traders.

(b) Weights for high- and low-quality forecaster in the Pred-Poll model.

(c) Average wealth for high- and low-quality traders in the Arti-PM.

Fig. 7. Traders' average historical Brier score, weight, and virtual wealth.

high-quality traders, with their superior performance, consistently dominate the Arti-PM's prediction outcomes, thereby reducing the influence of low-quality traders.

8. Conclusions and future research directions

We presented a novel artificial prediction market constructed with posters' comments to distill the wisdom of crowds within online communities. Compared to traditional opinion aggregation models, our experiments demonstrate that the proposed artificial prediction market excels in integrating crowd opinions from stock forums to predict price movements. Notably, our market outperforms all selected benchmark models in terms of predictability and profitability. We attribute this impressive performance to traders' wealth, which is a reliable proxy for their predictive ability and the extent of their efforts in updating their beliefs.

Our research yields three significant implications for researchers, investors, and corporations. First, the proposed solution offers profound insights into the development of artificial prediction markets and wisdom aggregation algorithms. Second, when comparing the proposed artificial prediction market with the prediction poll, the former can effectively aggregate the wisdom of crowds in online communities by assessing comment quality. In contrast, the latter struggles to aggregate the wisdom of crowds in online communities by assessing comment quality, especially in short-term predictions. Third, in other domains, such as geopolitics and crude oil trading, abundant online comments exist on platforms like Twitter or Facebook. Therefore, our artificial prediction market, established through online comments, presents an innovative tool for enhancing decision support across various domains, such as predicting geopolitical events, economic policies, and commodity price movements. Therefore, investors and companies can replicate our framework to design opinion mining systems for these scenarios.

While our paper demonstrated the prominent advantages of the proposed framework in extracting the wisdom of crowds within online communities for stock price movement predictions, there are several limitations and potential expansion. First, in online communities, interactions among posters may lead to systemic judgment or decision biases in the perspectives expressed in online comments. Therefore, future endeavors will explore incorporating machine learning models or reinforcement learning agents in artificial prediction markets that consist solely of artificial human traders. Doing so may alleviate the collective decision biases of traders, thereby enhancing market efficiency and predictive performance. Second, our study only focused on stock price movement predictions. Future research will explore the applicability and effectiveness of our method in other domains to validate its generalizability. Third, the traders in our experiments were drawn from a particular online community, only representing part of the population of investors or posters. Future studies will include more diverse participants, such as stock market analysts and financial news reporters, to improve the market performance.

CRediT authorship contribution statement

Li Dong: Writing – review & editing, Writing – original draft, Visualization, Software, Resources, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Haichao Zheng:** Writing – review & editing, Writing – original draft, Project administration, Methodology, Funding acquisition, Formal analysis, Conceptualization. **Liting Li:** Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Chunyu Zhou:** Methodology, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgements

This work was supported by National Natural Science Foundation of China [72071160].

References

[1] J. Gottschlich, O. Hinz, A decision support system for stock investment recommendations using collective wisdom, Decis. Support. Syst. 59 (2014) 52–62, https://doi.org/10.1016/j.dss.2013.10.005.
[2] S. Hill, N. Ready-Campbell, Expert stock picker: the wisdom of (experts in) crowds, Int. J. Electron. Commer. 15 (2011) 73–102, https://doi.org/10.2753/JEC1086-4415150304.
[3] Y. Zhang, L. Tao, Haze, investor attention and China's stock markets: evidence from internet stock forum, Financ. Res. Lett. 31 (2019), https://doi.org/10.1016/j.frl.2018.12.001.
[4] W. Tu, M. Yang, D.W. Cheung, N. Mamoulis, Investment recommendation by discovering high-quality opinions in investor based social networks, Inf. Syst. 78 (2018) 189–198, https://doi.org/10.1016/j.is.2018.02.011.
[5] X. Zhang, A.A. Ghorbani, An overview of online fake news: characterization, detection, and discussion, Inf. Process. Manag. 57 (2020) 102025, https://doi.org/10.1016/j.ipm.2019.03.004.
[6] J. Chang, W. Tu, C. Yu, C. Qin, Assessing dynamic qualities of investor sentiments for stock recommendation, Inf. Process. Manag. 58 (2021) 102452, https://doi.org/10.1016/j.ipm.2020.102452.
[7] M. Birjali, M. Kasri, A. Beni-Hssane, A comprehensive survey on sentiment analysis: approaches, challenges and trends, Knowl.-Based Syst. 226 (2021) 107134, https://doi.org/10.1016/j.knosys.2021.107134.
[8] R. Lukyanenko, J. Parsons, Y.F. Wiersma, The IQ of the crowd: understanding and improving information quality in structured user-generated content, Inf. Syst. Res. 25 (2014) 669–689, https://doi.org/10.1287/isre.2014.0537.
[9] K. Oliven, T.A. Rietz, Suckers are born but markets are made: individual rationality, arbitrage, and market efficiency on an electronic futures market, Manag. Sci. 50 (2004) 336–351, https://doi.org/10.1287/mnsc.1040.0191.
[10] J. Wolfers, E. Zitzewitz, Prediction markets, J. Econ. Perspect. 18 (2004) 107–126, https://doi.org/10.1257/0895330041371321.
[11] J.W. Goodell, S. Vähämaa, US presidential elections and implied volatility: the role of political uncertainty, J. Bank. Financ. 37 (2013) 1108–1117, https://doi.org/10.1016/j.jbankfin.2012.12.001.
[12] P. Atanasov, P. Rescober, E. Stone, S.A. Swift, E. Servan-Schreiber, P. Tetlock, L. Ungar, B. Mellers, Distilling the wisdom of crowds: prediction markets vs. prediction polls, Manag. Sci. 63 (2017) 691–706, https://doi.org/10.1287/mnsc.2015.2374.
[13] C. Borrero-Domínguez, T. Escobar-Rodríguez, Decision support systems in crowdfunding: a fuzzy cognitive maps (FCM) approach, Decis. Support. Syst. (2023) 114000, https://doi.org/10.1016/j.dss.2023.114000.
[14] J.E. Berg, T.A. Rietz, Prediction markets as decision support systems, Inf. Syst. Front. 5 (2003) 79–93, https://doi.org/10.1023/A:1022002107255.
[15] Y. Yu, W. Duan, Q. Cao, The impact of social and conventional media on firm equity value: a sentiment analysis approach, Decis. Support. Syst. 55 (2013) 919–926, https://doi.org/10.1016/j.dss.2012.12.028.
[16] T.H. Nguyen, K. Shirai, J. Velcin, Sentiment analysis on social media for stock movement prediction, Expert Syst. Appl. 42 (2015) 9603–9611, https://doi.org/10.1016/j.eswa.2015.07.052.
[17] A. Derakhshan, H. Beigy, Sentiment analysis on stock social media for stock price movement prediction, Eng. Appl. Artif. Intell. 85 (2019) 569–578, https://doi.org/10.1016/j.engappai.2019.07.002.
[18] M. Li, L. Chen, J. Zhao, Q. Li, Sentiment analysis of Chinese stock reviews based on BERT model, Appl. Intell. 51 (2021) 5016–5024, https://doi.org/10.1007/s10489-020-02101-8.
[19] Q. Wang, W. Xu, H. Zheng, Combining the wisdom of crowds and technical analysis for financial market prediction using deep random subspace ensembles, Neurocomputing 299 (2018) 51–61, https://doi.org/10.1016/j.neucom.2018.02.095.
[20] N. Oliveira, P. Cortez, N. Areal, Stock market sentiment lexicon acquisition using microblogging data and statistical measures, Decis. Support. Syst. 85 (2016) 62–73, https://doi.org/10.1016/j.dss.2016.02.013.
[21] S. Deng, A.P. Sinha, H. Zhao, Adapting sentiment lexicons to domain-specific social media texts, Decis. Support. Syst. 94 (2017) 65–76, https://doi.org/10.1016/j.dss.2016.11.001.
[22] N. Jing, Z. Wu, H. Wang, A hybrid model integrating deep learning with investor sentiment analysis for stock price prediction, Expert Syst. Appl. 178 (2021) 115019, https://doi.org/10.1016/j.eswa.2021.115019.
[23] Z. Jin, Y. Yang, Y. Liu, Stock closing price prediction based on sentiment analysis and LSTM, Neural Comput. & Applic. 32 (2020) 9713–9729, https://doi.org/10.1007/s00521-019-04504-2.
[24] R. Ren, D.D. Wu, T. Liu, Forecasting stock market movement direction using sentiment analysis and support vector machine, IEEE Syst. J. 13 (2018) 760–770, https://doi.org/10.1109/JSYST.2018.2794462.
[25] Q. Li, J. Wang, F. Wang, P. Li, L. Liu, Y. Chen, The role of social sentiment in stock markets: a view from joint effects of multiple information sources, Multimed. Tools Appl. 76 (2017) 12315–12345, https://doi.org/10.1007/s11042-016-3643-4.
[26] X. Song, G. Fan, M. Rao, Automatic CRP mapping using nonparametric machine learning approaches, IEEE Trans. Geosci. Remote Sens. 43 (2005) 888–897, https://doi.org/10.1109/TGRS.2005.844031.
[27] N. Oliveira, P. Cortez, N. Areal, The impact of microblogging data for stock market prediction: using Twitter to predict returns, volatility, trading volume and survey sentiment indices, Expert Syst. Appl. 73 (2017) 125–144, https://doi.org/10.1016/j.eswa.2016.12.036.
[28] R. Bar-Haim, E. Dinur, R. Feldman, M. Fresko, G. Goldstein, Identifying and following expert investors in stock microblogs, in: Proceedings of the Conference on Empirical Methods in Natural Language Processing, 2011, pp. 1310–1319. https://aclanthology.org/D11-1121.pdf.
[29] Y. Ruan, A. Durresi, L. Alfantoukh, Using Twitter trust network for stock market analysis, Knowl.-Based Syst. 145 (2018) 207–218, https://doi.org/10.1016/j.knosys.2018.01.016.
[30] Y. Luo, K. Liu, D.N. Davis, A multi-agent decision support system for stock trading, IEEE Netw. 16 (2002) 20–27, https://doi.org/10.1109/65.980541.
[31] D. Kanzari, Y.R.B. Said, A complex adaptive agent modeling to predict the stock market prices, Expert Syst. Appl. 222 (2023) 119783, https://doi.org/10.1016/j.eswa.2023.119783.
[32] Y. Huang, C. Zhou, K. Cui, X. Lu, A multi-agent reinforcement learning framework for optimizing financial trading strategies based on TimesNet, Expert Syst. Appl. 237 (2024) 121502, https://doi.org/10.1016/j.eswa.2023.121502.
[33] A. Barbu, N. Lay, S. Mannor, An introduction to artificial prediction markets for classification, J. Mach. Learn. Res. 13 (2012). https://www.jmlr.org/papers/volume13/barbu12a/barbu12a.pdf.
[34] F. Jahedpari, T. Rahwan, S. Hashemi, T.P. Michalak, M. De Vos, J. Padget, W. L. Woon, Online prediction via continuous artificial prediction markets, IEEE Intell. Syst. 32 (2017) 61–68, https://doi.org/10.1109/MIS.2017.12.
[35] J. Jumadinova, P. Dasgupta, A multi-agent system for analyzing the effect of information on prediction markets, Int. J. Intell. Syst. 26 (2011) 383–409, https://doi.org/10.1002/int.20471.
[36] J. Jumadinova, P. Dasgupta, Automated pricing in a multiagent prediction market using a partially observable stochastic game, ACM Trans. Intell. Syst. Technol. 6 (2015) 1–25, https://doi.org/10.1145/2700488.
[37] S. Lekwijit, D. Sutivong, Optimizing the liquidity parameter of logarithmic market scoring rules prediction markets, J. Model. Manag. 13 (2018) 736–754, https://doi.org/10.1108/JM2-06-2017-0066.
[38] N. Seong, K. Nam, Forecasting price movements of global financial indexes using complex quantitative financial networks, Knowl.-Based Syst. 235 (2022) 107608, https://doi.org/10.1016/j.knosys.2021.107608.
[39] R. Hanson, Logarithmic markets coring rules for modular combinatorial information aggregation, J. Predict. Mark. 1 (2007) 3–15, https://doi.org/10.5750/jpm.v1i1.417.
[40] A. Carvalho, A permissioned blockchain-based implementation of LMSR prediction markets, Decis. Support. Syst. 130 (2020) 113228, https://doi.org/10.1016/j.dss.2019.113228.
[41] T. Chakravorti, V. Singh, S. Rajmajer, M. McLaughlin, R. Fraleigh, C. Griffin, A. Kwasnica, D. Pennock, C.L. Giles, Artificial prediction markets present a novel opportunity for human-AI collaboration, arXiv Prepr. (2022) 16590, https://doi.org/10.48550/arXiv.2211.16590, arXiv:2211.
[42] G.W. Brier, Verification of forecasts expressed in terms of probability, Mon. Weather Rev. 78 (1950) 1–3, https://doi.org/10.1175/1520-0493(1950)078<0001:VOFEIT>2.0.CO;2.
[43] R. Buhagiar, D. Cortis, P.W. Newall, Why do some soccer bettors lose more money than others? J. Behav. Exp. Financ. 18 (2018) 85–93, https://doi.org/10.1016/j.jbef.2018.01.010.
[44] J. Grunert, L. Norden, M. Weber, The role of non-financial factors in internal credit ratings, J. Bank. Financ. 29 (2005) 509–531, https://doi.org/10.1016/j.jbankfin.2004.05.017.
[45] H. Zou, T. Hastie, Regularization and variable selection via the elastic net, J. R. Stat. Soc. Ser. B Stat Methodol. 67 (2005) 301–320, https://doi.org/10.1111/j.1467-9868.2005.00503.x.
[46] F. Mai, S. Tian, C. Lee, L. Ma, Deep learning models for bankruptcy prediction using textual disclosures, Eur. J. Oper. Res. 274 (2019) 743–758, https://doi.org/10.1016/j.ejor.2018.10.024.
[47] J. Du, Mean–variance portfolio optimization with deep learning based-forecasts for cointegrated stocks, Expert Syst. Appl. 201 (2022) 117005, https://doi.org/10.1016/j.eswa.2022.117005.
[48] H.M. Markowits, Portfolio selection, J. Financ. 7 (1952) 71–91, https://doi.org/10.2307/2975974.

Li Dong is a Ph.D. candidate of technology economics and management at the School of Management Science and Engineering, the Southwestern University of Finance and Economics, China. His research interests lie primarily in human-machine hybrid prediction system and financial intelligence. He has published in the *Electronic Commerce Research and Applications*.

Haichao Zheng is a professor at the School of Management Science and Engineering, the Southwestern University of Finance and Economics, China. He received his Ph.D. in management science and engineering from Nankai University. He was a visiting scholar at the University of Minnesota Duluth from 2009 to 2010. His research interests include crowdsourcing, crowdfunding, and human-machine hybrid system. He has published in

the *MIS Quarterly, Decision Support Systems, Information Systems Journal, European Journal of Information Systems, Information and Management, International Journal of Electronic Commerce, Electronic Commerce Research*, and other publications.

Liting Li is a lecturer at the College of Economics and Management, the Taiyuan University of Technology, China. Her research interests are in the area of crowdsourcing and human-machine hybrid system. She has published paper in the *MIS Quarterly, Electronic Commerce Research, Electronic Commerce Research and Applications, Journal of Global Information Managemennt*, and others publications.

Chunyu Zhou is a quantitative researcher in the finance industry at Qube Research and Technologies, China. She obtained her master's degree in computer science from the University of Hong Kong and her bachelor's degree in financial engineering from the Southwestern University of Finance and Economics. Her research interests are in the area of machine learning, quantitative finance, and econometrics.